Inductive Reasoning Renewed: A Reply to Commentators

Joachim I. Krueger[1]

David J. Grüning[2,3]

Patrick Heck[4]

David Freestone[5]

[1] Brown University

[2] Heidelberg University

[3] GESIS – Leibniz Institute for the Social Sciences

[4] Consumer Financial Protection Bureau (CFPB)

[5] Equip Health, Inc

Correspondence:

Joachim I. Krueger

Department of Cognitive, Linguistic & Psychological Sciences

Brown University, Box 1820

190 Thayer Street

Providence, RI 02912

Phone: (401) 863-2503

E-mail: Joachim_Krueger@Brown.edu

Abstract

The IRM provides a simple and flexible framework for the study of the social-perceptual phenomena of self-enhancement, intergroup accentuation, ingroup-favoritism, and differential accuracy of group perception, as well as the interdependencies of these phenomena. The commentaries on the target article (Krueger and colleagues) point to opportunities to further develop the model and to situate it in the context of contemporary theory and research in social and cognitive psychology. We address several critical issues of concern in this rejoinder. We organize our arguments along four themes of discussion: the model's coherence, validity, sufficiency, and its timeliness. We close by suggesting that the IRM, beyond its proper domain of applicability represents a way of thinking about social perception characterized by parsimonious assumptions and precise predictions.

Keywords: inductive reasoning, social perception, modeling

We are a community in the thinnest sense, and that is, a community of disputatious individuals.
– Richard Shweder, commenting of the Kalven Report, The Chicago Maroon, May 14, 2014

Having added another moderator variable, you are no closer to understanding complexity.
– Hoca Camide (undated)

With our commentators, we participate in an intellectual community seeking advances in psychological science and doing so in the face of constructive criticism. We are indebted to our colleagues for sharing interesting and provocative ideas and suggestions, and for showing us where the IRM falls short, needs revision, and deserves further development. The generally positive tenor is more muted in the remarks of Sedikides and Skowronski who feel that the IRM is a return to a kind of theorizing that flourished in the 1950s and 1960s and that has since been overcome. Hoorens and colleagues also find much to question and critique. So, was it worth the trouble developing the IRM? We conclude with a cautious and hopeful "yes," and will explain our reasoning in this rejoinder. We attempt to steer clear of defensive dissonance reduction, but, being human, we cannot guarantee complete success on this score, only good intent.

Reading critical commentary can stimulate dissonance and the attendant temptation to justify one's efforts. The IRM is the results of decades of work, and for the first author in particular, a personal milestone of achievement and closure. This closure, we hope, will open new doors of exploration instead of being the end of a line. Several commentators see interesting and useful directions in this regard. As a model of parsimony, the IRM is not meant to foreclose on out-of-model exploration, but to encourage it. The IRM, as we did not tire to point out in the target article, is meant to present a baseline of expectation and a roadmap for the joint study of several social-judgmental phenomena of enduring interest. So yes, bring it on! Empty the bag of tricks and pour out your favorite moderator variables, boundary conditions, anomalous cases, and alternative metrics! Many of the variables the commentators have proposed as worthy of further study hold promise and many will likely offer increments in explained variance. *Caveat emptor*, however, because a glut of formerly overlooked variables may come back to haunt the diligent researcher with a nostalgia for a parsimonious model, a model like, say, the IRM.

One of us [JIK] remembers an off-the-cuff comment a senior colleague, Donelson (Don) Forsyth, a distinguished researcher of small group dynamics and a pioneer of moral psychology (e.g., Forsyth, 1980), made a quarter century ago at a small group conference in the Alpine foothills. "Parsimony," Don declared, "is a false god." JIK was too stunned at the time to counter with a viable retort such as "Tell that to a monotheist!" We now know that parsimony is not a god, but a heuristic, and a useful one. Every scientific model worth its salt must strike a balance between limiting the number of assumptions and parameters (parsimony) and including enough of them to tame noise, uncertainty, and the unexplained (fit). To promote parsimony is to tolerate uncertainty (Gigerenzer et al., 2022). It is to protect oneself from overfitting the data. Logically, it is possible for there to be (ironic) cases of too much parsimony, but who can think of a good example?

The commentaries raise issues, concerns, and possibilities that, we believe, fall into four general rubrics: coherence, validity, sufficiency, and timeliness. The dimmest view one might take on the IRM is that it is incoherent, not falsifiable (or has been falsified already – and never mind the incoherence of that), starved for contents (underspecified), and anachronistic. Let us consider these four types of concern.

Coherence

A theory that makes contradictory assumptions is in trouble, a view that can be traced to Spinoza (Young, 2024). Hoorens and colleagues note that the IRM assumes differential projection, that is, greater projection to ingroups than to outgroups, while also allowing that evidence gathered from observing other individuals supports inductive inferences about the properties of these others. Clearly, there are more opportunities to observe ingroup than outgroup members, and yet the IRM seems to assume that perceivers project both more (the social distance effect) and less (the social learning effect) to ingroups than the outgroups. This would be a contradiction – if it were so.

The misunderstanding, we submit, arises from the mistaken idea that all social projection is, by definition, an egocentric bias. However, ingroup projection, that is, r_{SI}, may – and should – assume positive values when perceivers learn about the behaviors or traits of ingroup members and find that the similarities dominate the dissimilarities. This should be especially so in real, that is, non-minimal, groups, whenever true interpersonal similarities contribute to social categorization (Ames, 2004). There

is a long history of attempts to separate egocentric overprojection from reality-based simple projection. These efforts led to the proposal and use of various discrepancy scores, most of which suggested that projection is more likely to be too strong than not strong enough (see Krueger, 1998a, for a review). Yet, these discrepancy-score measures never resulted in reification of the concept (Krueger et al., 2017a). There are no grounds for concluding that overprojection, or projective bias, is a reliable psychological phenomenon able to stand on its own, and be treated as a trait-like individual difference. As these efforts to isolate projective bias petered out, we came to settle on the correlation r_{SI} as a serviceable statistical index for the inclination to expect and see self-other similarities, true or false (Krueger, 2008).

A careful look at the meta-analytic findings reported in Robbins and Krueger (2005) shows that there is no contradiction. Perceivers project more to ingroups than to outgroups, and this differential is greater for minimal groups than for real, that is, non-minimal, groups. People project less to real ingroups than to minimal ingroups because they have learned from observation (or self-stereotyping) that some fellow ingroup members feel or behave differently than they themselves do. They project more to real outgroups than to minimal outgroups because they have learned from observation (or stereotyping) that even outgroupers share some similarities with them (Krueger & DiDonato, 2008), thanks to their shared humanity or being subsumed in other superordinate categories (Krueger & Clement, 1996; see also Neel & Hehmann). This is all as it should be, showing the complementary effects of social projection and non-egocentric social induction.

Our reluctance to grant projective bias, or "overprojection," ontological status extends to the concept of self-enhancement. Self-enhancement, typically defined as a self-image more positive than warranted, implies by its very name a positive *shift*, and thus a driven discrepancy between how things are perceived and how they are. The same types of discrepancy score used to capture overprojection have been used to capture self-enhancement, that is, difference scores, regression residuals, and their various derivatives (Krueger & Wright, 2011). Of course, some authors still insist on measuring self-enhancement directly by asking respondents to rate themselves relative to the average person on a scale from, say +5 (much better) to -5 (much worse). This measurement strategy totally confounds legitimate differences in the positivity of self-images with irrational, motivated, or accidental distortions.

The IRM respects self-enhancement as a measurable, predictable, and understandable phenomenon, but it does not treat it as a thing or a property of person or mind. Instead, the IRM models self-enhancement as an outcome of a positive self-image and less-than-perfect social projection. We should note a complication: If self-enhancement is estimated as r_{SD} - r_{SD} x r_{SI}, we ought not conclude that r_{SD} x r_{SI} is the self-image perceivers *should* have. This would be incoherent as r_{SD} appears twice. We might then iterate the process and claim that self-enhancement is given by r_{SD} x r_{SI}, - r_{SD} x r_{SI} x r_{SI}, and so forth. Instead, the estimate r_{SD} x r_{SI} reflects the implied positivity of the average ingroup member and the difference from r_{SD} might reflect other-diminishment instead of self-enhancement. In keeping with standard usage, however, we refer to the difference between the positivity of the self-image and the positivity of the group image as self-enhancement. Importantly, this difference remains an estimate of bias. It does not separate out the error component included in this bias (see Heck & Krueger, 2015, or Heck & Krueger, 2020, for efforts to achieve such a separation).

The IRM, Hoorens and colleagues and Alves claim, contains circularities. They question whether positive self-views can be achieved through inductive reasoning when these self-views themselves serve as input for further inductive reasoning. More specifically, they charge, the IRM's assumption that self-views are mainly positive is made with reference to the literature of self-enhancement. We see no circularity here. It is uncontested that most people report positive self-views (Sedikides et al., 2021), and there is no compelling reason why any self-positivity must necessarily be regarded as a sign of bias, or, worse, an error. The modest contribution made by the IRM is to estimate a self-other difference in positivity that might also be seen as a form of other-diminishment. The intent is to preserve the integrity of positive self-views; they are not self-enhancing by definition (Krueger et al., 2017b). When we observe that most people report themselves to be happy (Diener et al., 2018; Wedell & Parducci, 1988) and to have high self-esteem (Baumeister et al., 2003), we can refer to many processes aside from self-other comparisons that make it so. Intriguingly, most people believe their own self-esteem is higher than the self-esteem of the average person (Krueger, 1998a). This then may be self-enhancing or other-diminishing. In sum, we submit that the IRM is as coherent as other models (see Glöckner & Fiedler).

Validity

If we can accept the stylized empirical facts of positive self-images and differential projection to ingroups and outgroups, and if we accept the correlations among these variables as legitimate statistical estimates of association, we need to address questions about the IRM's testability and falsifiability. The model's machinery is a matter of mathematical implication, but one might question the stylized empirical facts, by suggesting with Hoorens and colleagues that self-images are not really positive (respondents are lying to us or to themselves), by suggesting with Thielmann that social projection fades away in the domain of personality traits, or that projection is not differential when outgroups that are similar to the self are being considered.

Once its predictions are made, the IRM because of its parsimony, leaves room for "something else going on." It invites theorists to generate testable hypotheses going beyond the model's basic predictions. It is not the task of the IRM (or any model) to account for all conceivable contexts. For example, since self-images are more closely tied to moral than intellectual traits, we can say we have found an important moderator (Strohminger & Nichols, 2014). Critically, however, such a moderator does not affect the structure of the IRM, and it does not falsify it. Knowing, or hypothesizing, that the moral domain affects both self-positivity and ingroup projection, the IRM shows that the effects of these two types of input work in opposite directions, and the model makes specific predictions contingent on one of the two effects being stronger than the other.

Koch and colleagues make several recommendations for the validation of the IRM. They point to the possibility of experiments manipulating the values of the model's input parameters. If, for example, a person's self-image were nudged to be more positive, the IRM would predict increases in self-enhancement and ingroup-favoritism and no change in intergoup accentuation. Such results would corroborate the model's internal validity. A comparison between the IRM and the cognitive-ecological model, CEM (Koch et al., 2024) would add little, however, to a comprehensive appraisal of either model. These models are rather orthogonal to each other. Whereas the CEM focuses on the effects of judgment order, the IRM is mute on this issue. At minimum, it would be encouraging to see that the IRM's predictions are not heavily moderated by judgment order, but we do not see how the predictions of the two models can be pitted against each other. By contrast, we think that the probabilistic reformulation and expansion of the IRM by Johnson is a good example of how questions of internal

validity can be addressed. It appears that the IRM has survived the probabilistic challenges well. Its relation to process models of cognition (e.g., Bayesian or connectionist; Glöckner & Fiedler) awaits further exploration. The development of an open-access infrastructure (also suggested by Glöckner and Fiedler) would be an excellent initiative to further advance, test, and adapt the IRM.

Neel and Hehmann explore the IRM's internal validity by considering its mathematical structure. Following their lead, we see a new opportunity to address the perennial question of whether perceptions of self-other similarity reflect social projection or self-stereotyping or a mixture of both (Cadinu & Rothbart, 1996). In the past, we have emphasized the primacy of the self-perception (e.g., Krueger, 2003). The IRM offers a new type of test. The model predicts ingroup positivity as $r_{ID} = r_{SD} \times r_{SI}$, where r_{SI} is an index of perceived self-group similarity. If r_{SI} were to reflect self-stereotyping, then this self-stereotyping should drive the positivity of the self-image, such that $r_{SD} = r_{ID} \times r_{SI}$. Projection theory, however, assumes that the positivity of the self-image is an autonomous input, which drives perceptions of group positivity and not *vice versa*. Thus, projection theory (and with it the IRM) predicts that $r_{SD} > r_{SI} \times r_{ID}$

As to external validity, Koch and colleagues recommend the study of holistic variables. In this regard, we note the findings of Rivera et al. (2024), brought to our attention by Ullrich and Sebben, which suggest that individual self-esteem is more predictive of ingroup than of outgroup evaluations. Likewise, as Neel and Hehmann propose, the question of having multiple ingroups or multiple outgroups deserves attention. As these suggestions point to fertile research ground, we think that the question of whether the IRM can be falsified is not instructive. Critical rationalism, as the joke goes, has been falsified, in contrast to Bayesianism, which does not deal in certainties. Most theories do not die violently, they fade away (Lakatos, 1971). They are laid to rest one funeral at a time, as Max Planck quipped. As psychologists increasingly favor "modeling" over "theorizing," they have moved to protect their ideas from being falsified. A model may not fit the data as well as an alternative model does (Vrieze, 2012), but his hardly counts as falsification. Indeed, the better-fitting model may be overfitting the data (Gigerenzer, 2020). The IRM underfits the data by design. It offers theoreticians and modelers an opportunity to hypothesize and show what else might be going on.

We have presented the IRM's ability to make specific, non-nil, point predictions as an advance, following Meehl's (1967; 1978; see Trout, 2004, for an introduction and review) suggestions regarding strong significance testing. Point predictions, be they nil hypotheses (i.e., the true effect is 0) or non-nil hypotheses (e.g., the size of the self-enhancement effect is .25) eventually surrender to big data yielding small *p* values (Krueger & Heck, 2017, 2018). If all point predictions are ultimately shown to be false, why bother testing them? We take the position that nil hypotheses and non-nil hypotheses are qualitatively different in that the former may be entertained on the view that there is only stochastic noise. The latter may be entertained on the view that there are systematic and non-trivial associations between variables. If, for example, we hypothesize with Thielmann that some traits do not lend themselves to being projected, and we find that a projection correlation of .2 to be significant (thanks to big data), we might shrug off this small effect. By contrast, if we pose an IRM-informed hypothesis of .25 for self-enhancement only to then find an empirical effect of .35 (with $p < .05$ for the difference), we suspect that something else is going on. If this difference were not significant, we would not revert to the agnostic belief that nothing at all is going on. In short, traditional significance testing against 0 guides inferences regarding something vs. nothing (Krueger & Heck, 2019), and it encourages dichotomous decision making (Gelman & Stern, 2006; Greenland, 2017). Strong significance testing, by contrast, suggests that a theory be refined, not abandoned.

The recovery of Meehl's proposal to test non-nil hypotheses struck a discordant note with some commentators. We observe, however, that strong hypothesis testing is already part of the fabric of psychological science. Meta-analyses, for example, now show whether effect sizes obtained in individual studies lie outside the confidence interval bracketing the grand average effect size – which may not be zero. If only a few effect sizes lie outside that interval, they might be regarded as anomalies and not as falsifiers (see, for example, Genschow et al., 2023). The evidence for differentiated differential projection, that is, larger ingroup-outgroup differences in projection between minimal than between real groups, can also indicate strong hypothesis testing. Taking the average projection effect for a minimal ingroup as a point prediction when testing for projection in real groups, significance tests do not treat "nothing" as the baseline. The difference is one of perception rather than substance. The traditional interpretation would be that a zero difference between projection to real groups and

projection to minimal groups is a nil hypothesis. If, however, we argue that projection to minimal ingroups is a theory-guided prediction and if we accept the empirically supported strength of projection to a minimal ingroup as a point prediction, we can ask whether projection to real groups significantly differs from *that*.

The IRM not only generates point-predictions for derivative social-perceptual phenomena, but it also generates ranges of possible values. Following the pioneering physiologist, psychologist, and statistician Johannes von Kries (1988), we refer to this range of possible values as the *Spielraum*, literally the room available for play (see also Meehl's, 1967, concept of *tolerance range*). Consider ingroup favoritism. With ingroup projection, r_{SI}, and outgroup projection, r_{SO}, being respectively .5 and 0, and with self-positivity, r_{SD}, being .5, ingroup positivity can range from +1 to -.5, and outgroup positivity can range from +.87 to -.87. This implies a maximum difference between the two group positivities of +1.87 and a minimum difference of -1.37. Hence, the *Spielraum* is +3.23. The IRM cannot be falsified by empirical values lying outside of this range. Some things cannot be so. A model's ability to make point predictions, and its ability to show the boundaries of possible results is an advance, as acknowledged by many commentators, including Johnson who introduces probabilistic refinements. If Sedikides and Skowronski think that ordinal predictions *can* be just as valuable they are correct, if a bit short-sighted. Consider the predictions made in early work on judgment (Tversky & Kahneman, 1974) and choice (Kahneman & Tversky, 1979). These predictions were qualitative, that is, they were ordinal and directional. The heuristics that were uncovered were stimulating and powerful, but the underlying theory was weak (Gigerenzer & Gaissmaier, 2011; Regenwetter et al., 2022). The IRM, by contrast, is theoretically grounded and it enables significance testing in its strong form.

To further evaluate the model's merits, we now look beyond questions of validity. Having made the case for the model's coherence, we turn to the issue of parsimony, asking whether the IRM's insufficiencies spoil its usefulness.

Sufficiency

The IRM does not aspire to be a complete model of social perception. Myriad omitted context variables suggest themselves for consideration. To clarify how the IRM is positioned in this thicket of variables, we address four questions: First, is the IRM flawed because it produces predictions without

recourse to social comparisons? Second, is the IRM limited because it does not address the selection of traits and because it says nothing about judgments of single traits? Third, is the IRM insufficient because it derives measures of judgmental accuracy from collective self-ratings? Fourth, is the IRM hampered by providing its best fit in information-sparse environments?

First, where did all the comparisons go? Many psychophysicists from Weber to Helson to Parducci held that judgment is necessarily comparative. Indeed, judgments are not made in a vacuum. Humans store memories of sampled experience so that any judgment of a particular stimulus is likely to be affected by its psychological context. Judgments made with the usage of verbal labels such as "heavy," "tall," or "beautiful," may have to answer to the question "compared to what" (Dawes, 1988)? We propose, however, that judgments do not *necessarily* reflect either an attraction to or repulsion from some referent stimulus (Mussweiler, 2003). Instead, we maintain that some judgments are absolute, being dependent on sensory systems that constrain what can be felt. These systems establish a *Spielraum* for perception, a range of what is possible to be sensed. When we look at the sun, we know it is bright. No comparison with the moon's gentle light is needed. When we mourn the loss of a friend, we need not note that losing a pet is less painful. When we are exposed to very low-intensity stimuli, the question is how much stimulation is sufficient for the person to feel that there is something as opposed to nothing (Swets et al., 2000; Weber, 1850).[1] Likewise, cognitive-sampling models do not assume that judgments of frequency require alternate categories to be sampled (Fiedler & Juslin, 2006; Fiedler, Juslin, & Denrell, 2023; Galesic et al., 2012). Some frequency judgments can stand on their own.

Many prominent theories of social judgment take the ubiquity of comparisons for granted, and they build this assumption into their measurement operations. Tajfel's (1969) work on accentuation theory, which has been particularly inspiring to us (Krueger et al., 1989), explicitly addressed between-category contrasts and within-category assimilations. Likewise, Tajfel's (1970) work on seeing between-group differences where there are none, has been a guiding light (DiDonato et al., 2011). It is

[1] We regard judgments at the threshold as absolute judgments. To say that such judgments are in fact comparative, if *nothing* serves as a referent for the detection of *something*, is to say that all judgments are comparative by definition.

because of this prevailing consensus that between-group comparisons are essential that the IRM's ability to reproduce these effects without invoking comparisons is so important (see also Gaertner et al., 2006).

Much of the research on the Better-Than-Average effect as the signature of self-enhancement bias places self-other comparisons at the core of self-knowledge. The bias is thought to be revealed by positive values on a comparative +5 to -5 scale. Years ago, Klar and Giladi (1999) showed that direct comparative self-other ratings were predicted by absolute self-ratings but not by absolute other-ratings. The confounded nature of direct comparative ratings should have been obvious from the outset, and if not, the Klar-and-Giladi findings should have ended their use (Krueger et al., 2013). But this is not what happened. Researchers love their comparisons (Zell et al., 2020).

Intra-person comparisons can also come into play (Heckhausen & Krueger, 1993; Krueger & Heckhausen, 1993). Many varieties of emotion and motivation pick up on a sensitivity to change, much as psychophysical judgments do (Weber, 1850). Departures from equilibrium may be corrected or the person adapts to a new context and level of stimulation (Helson, 1964). Such intra-perceiver comparisons provide input for self-judgment, but they occur before the processes (specifically, projection) and outcomes predicted by the IRM are set in motion. They therefore lie outside of the model's operations. A final comment is of interest here with regard to the question of whether social projection itself is a matter of comparative judgment. Might one not see strong projection as an assimilative judgment and counter-projection as a contrastive judgment, that is, a judgment of false uniqueness (as suggested by Hoorens and colleagues)? As to the former, we observe that sampling models are sufficient to describe projection (Dawes, 1989). As to the latter, counter-projection makes no sense from a sampling perspective, unless one admits additional prior beliefs about the distribution of traits in a group. But then, the perception of one's uniqueness would not be false, but true. Indeed, evidence of false uniqueness is truly unique (Krueger et al., 2006).

Although comparisons spring to mind easily, it is hard to image a perceptual world exclusively dependent on comparisons. A large number of pairwise similarity or dissimilarity judgments can be assembled and a distance map can be constructed, but the entire cloud of objects floats in an unnamed space unless there is a metric investing the ordinal (dis)similarities with an absolute meaning (Hebart et

al, 2020). If, for example, all recruits are compared, pair by pair, to determine their relative height, we still don't know who will look good in a 44 regular.

The second question we address is whether the IRM is hampered by its silence concerning which traits are to be selected for measurement. We have assumed that a broad range of traits be assembled (see also Judd & Park, 1993; Westfall et al., 2014), ranging from highly negative to highly positive ones. Biesanz agrees, but also asks if normative (that is, averaged across respondents) trait-desirability may be used in lieu of each respondent's own desirability ratings. This would not make much of a difference to the IRM, although it would make self-images slightly less positive (Krueger, 1998b; Sinha & Krueger, 1998) and thereby reduce self-enhancement and ingroup-favoritism.

A set of randomly distributed traits provides a good baseline. Departing from that, special subsets or even anomalous individual traits may be identified to show that the IRM does not explain everything. What if only positive traits were used? Ullrich and Sebben invite us to consider a pattern of correlations consistent with the IRM and to also imagine that ratings for the ingroup are lower than ratings the ingroup. Hence, we'd see ingroup favoritism through the correlational lens while seeing outgroup favoritism through the mean-level lens (see also Neel & Hehmann). This is an intriguing arrangement. Although it is possible to occur, it is empirically unlikely. Indeed, this thought experiment reinforces the recommendation to use a broad and representative sample of traits when working with the IRM. As the correlational metric is only one way of looking at statistical associations and differences, it is good practice to explore the data in multiple ways (Cadinu & Rothbart, 1996; Cronbach, 1955; Krueger, 2009). Ullrich and Sebben also note that trait ascriptions may not need to be independent of one another. Their dominant-submissive example is instructive. It captures an interdependence of self-other perception that lies beyond the reach of the IRM.

A representative sample of traits will include traits across the so-called big-five domains (or domains specified by other taxonomies). Thielmann questions the wisdom of representative sampling, noting that value-related domains, such as agreeableness or openness to experience, show strong projection effects in correlations over respondents, whereas other domains (extroversion, conscientiousness, neuroticism) do not. As an aside, we think that the dimension of neuroticism vs. emotional stability is value-laden. Thielmann thus clarifies an important boundary condition of the

IRM. As a type of Bayesian induction under uncertainty (Dawes, 1989), ingroup projection requires ignorance of the distribution of scores in the absence of sampling. Such ignorance is a feature of nature before the psychometricians enter the scene. Psychometric measurement and scale design seek to reproduce the distributions typical in nature, namely Gaussian ones. The science of biostatistics is staked on the normal distributions found to characterize most interval-scaled natural features, such as an organism's height or weight (Stigler, 1999). The great Belgian mathematician Adolphe Quetelet (1871) was so impressed by the normal distributions around physical-property averages that he exulted the average man as a beautiful prototype. The average man, in his view, is a Platonic ideal. Deviations from this ideal reflect the flickers of a flame. Statistically, they are modeled as error.

Psychometricians do not always find normally distributed measurables in their domain of study. To overcome this limitation, they often construct measurement devices to force normal distributions into being. Intelligence tests, for example, are normed to yield a mean of 100 and most of them yield a standard deviation of 15 (Kaufman, 2000). Many personality traits are no different. Are 20% of people selfish, or is it 50%? It depends on the threshold that is applied. Whereas stable individual differences surely exist in nature, the shape of the distribution of scores and its mean depend on the measurement items comprised by the scale. To avoid the complications that arise from this, researchers often insist on standardizing personality measures to keep their eyes trained on individual (or perhaps group) differences (Fritz Ostendorf, personal communication, ca. 1990). The question of whether people are on average more agreeable than they are extraverted is indeterminate in this context. The results is that respondents who understand they are participating in a measurement exercise where the average and the median score are the same have no room to project. The extrovert has to agree with the introvert that 50% of people are extroverts – by virtue of scale construction.

It is impressive that even in these trait domains some projection is observed. If Thielmann suggests $r = .2$ as a minimum required effect size for social projection, it should be noted that such an effect is roughly equivalent to $d = .4$, which is more than a third of a standard deviation (Funder & Ozer, 2019; Lakens, 2022). Greenwald et al., (2015) observed *average* effects of around $r = .2$ in a different domain, and they defended their theoretical and social importance. Incidentally, Thielmann

also finds projection to outgroups, a result which is at variance with the view that there is little projection overall.

Thielmann, like Ullrich and Sebben, ask us to consider the IRM's applicability to individual traits. For individual traits, the IRM and other correlational analyses do not work well (Hoch, 1987). The IRM was designed to allow person-specific assessment. Within-trait correlations or mean differences only speak to group effects. *Nota bene*: the IRM allows the study of group effects, and it does so without sacrificing the person-centered psychological interest (Hoch, 1987).

As an illustration of the limits of within-trait analyses, consider the phenomenon of intergroup accentuation. This phenomenon is studied effectively with a focus on mean differences (Krueger, 1992; see Rothermund & Deutsch, 2024, for recent research). Of course, a mean difference can be converted to a point-biserial correlation coefficient, but this correlation do not deliver much of interest in the IRM sense. If as $r_{IO} = 0$, ingroup projection or outgroup projection or both are nil, or the responses are just noisy. A finding $r_{IO} = -1.0$ makes no sense, and a finding of $r_{IO} = +1.0$ would suggest the presence of some odd individual difference in response bias. Again, a within-trait correlation is not a suitable index of accentuation.

The third question is whether the IRM's approach to the assessment of judgmental accuracy is warranted, given that a group's aggregated self-ratings may be biased, especially in a positive direction (Biesanz). We regard a perceiver's group judgments as accurate inasmuch they are positively correlated with the average self-judgments of group members over traits. This is a legitimate approach to accuracy if its definition and its limitations are understood. Suppose group X objectively comprises more negative traits than does group Y. If members of group Y rate group X more negatively than members of group X, thanks to positivity biases, rate their own group, members of group X would assuredly claim that they are being unfairly stereotyped by members of group Y. The impasse, or misunderstanding if you will, could be resolved if Omniscient Jones intervened, admonishing members of group X to adopt a more negative and more accurate self-view. This would not sit well with group X. With Omniscient Jones only rarely making an appearance in the world of social perception, aggregated self-reports may serve as an acceptable proxy (Terracciano et al. 2005; see Krueger & Wright, 2006, for a commentary). Inasmuch as each respondent's self-ratings are a partial expression of what the group "is really like,"

aggregating these ratings capitalizes on the accuracy-yielding wisdom of the crowd (Kameda et al., 2022; Krueger & Chen, 2014). Did we not accept aggregated self-reports as a proxy of truth, we would have to say, like Judd and Park (1993, p. 114) did with a stroke of a pen that "for the purposes of this discussion, we assume that a valid criterion is available." But we don't have that sort of external criterion for many questions of interest (see Jussim, 2012, for a review of reality-grounded accuracy in social judgment).

The fourth question is whether the IRM, though working well for minimal groups, is useful in the information-rich environment of real social groups (Alves). Our argument for the usefulness of the IRM for real social groups – beyond the points already made about information integration, projection and projective bias – focuses on the idea that the minimal group paradigm, MGP, continues to be instructive today, and more so than typically realized. After it had been foreshadowed for years (e.g., by Sherif & Sherif, 1963), the MGP achieved its present form in the work of Henri Tajfel (1970, see also Tajfel et al., 1971) and Jaap Rabbie (Rabbie & Horwitz, 1969; see Brown, 2020, for an historical overview). The MGP was a breakthrough in experimental social psychology, allowing researchers to study phenomena such as accentuation, discrimination, prejudice, and social identity formation in the purified ecology of the laboratory. As all social information besides classification itself was removed, minimal groups were as radical an innovation as Evangelista Torricelli's creation of a physical vacuum in 1643. This analogy was not lost on Tajfel (1972). The social vacuum was so perfect that Tajfel had no theory to explain the ingroup-favoritism he found. Only years later did he propose social identity theory to account for the findings (Tajfel & Turner, 1979). Notably, this theory assumed that ingroup-favoritism serves to prop up self-esteem, an assumption the IRM inverts. Rabbie and Horwitz (1969) found ingroup favoritism only when their participants shared a common fate of either all winning or not winning a coveted prize. The concept of common fate invoked a perceptual effect described by the early *Gestalt* theorists (Wertheimer, 1923). In later work, Rabbie incorporated ideas related to inductive reasoning, such that research participants were thought to favor ingroup members on the assumption that these ingroup members would return the favor (Rabbie et al., 1989). In other words, individuals were thought to project their discriminatory behavior on fellow ingroup members. This hypothesis has since been corroborated (Krueger et al., 2012; Yamagishi & Kiyonari, 2000). Models of inductive

reasoning, expected reciprocity, and biased memory (Rothbart, 1981) began to address the Achilles heel of the early work, namely the conflation of the *explanans* with the *explanandum* (Alves). Having found ingroup favoritism, the early investigators had delivered a proof of concept, but they were unable to explain the phenomenon in psychological terms without leaning on the phenomenon itself for explanatory assistance. The IRM, along with other contemporary models, addresses this problem.

The social world is characterized by much fluidity. Groups form and dissolve; many social categories are invented as a response to a unique historical moment, enjoy a honeymoon of high fashion, only to be replaced by an alternate set of categories soon after. The sampling, observing, and remembering individual may try to keep track of the stream of information, group it reliably, and form rational and accurate inferences, but it is much to ask in a volatile environment. By contrast, the self, or rather self-related information and knowledge is always available. Wherever you go, as the saying goes, there you are. The self-concept is a rock in the surf of social information. We should not be surprised when people use it, and even overuse it, when seeking to understand the buzzing confusion of social reality. The study of psychological processes in the laboratory remains, in our view, an indispensable tool in a science of mind. The exploration of variation of these processes in the wild is interesting and socially desirable, but it is not essential (Krueger, 2023; but see Kroupin et al., 2024).[2]

Timeliness

Most commentators see the IRM as a promising advance in the study of social perception. Sedikides and Skowronski, however, see the IRM as a step back, and they refer to their own Egocentric Tactician Model, ETM, as a prototype of how to move social perception research forward. We agree that the ETM is ambitious, innovative, and integrative, while maintaining that the same can be said for the IRM. We now review the criticisms Sedikides and Skowronski level at the IRM, and then ask whether the ETM meets these same critical standards. We will, as some commentators would have it (e.g., Alves), engage in social comparisons because we must.

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[2] Reasonable people disagree on this point. Some see unity where others see multiplicity. This core difference in perspective runs through the intellectual history of the West, after it was first seen in the works of Parmenides (unity) and Empedocles (diversity) (Russell,1946).

Our critics feel that the IRM is throwback to the descriptive type of modeling, which held sway half a century ago, modeling that used some mathematical tools, but failed to theorize about and explore mental concepts, processes, and mechanisms. This type of modeling, they claim, lacks depth; it promotes a "shallow psychology" (Dawes, 1976). Our short answer to this charge is "Yes." We see value in disciplined descriptive modeling and prediction (Krueger, 2020). Further, we do not make a sharp distinction between description-prediction on the one side of the coin and process-oriented research on the other. We hold the research on the origins of social projection and positive self-views in high regard. It is not the task of the IRM to contribute further to this literature. Its task is to raise the bar for additional and unique processes invoked to explain self-enhancement, ingroup-favoritism, intergroup accentuation, and differential accuracy. With the IRM, we show that these phenomena *may* be derivatives. The model does not claim that these phenomena *must* be reductive. Something else might be going on.

A longer answer emerges from a consideration of the proposed alternatives. Our critics fault the IRM for neglecting complexity. The world's climate and the human brain are highly complex systems, so complex that only a model that as complex as its target can fully capture this target. We all agree that some reduction of the number of parameters is required for useful scientific work. The way the IRM does this, our critics assert, is Newtonian, using a paradigm that has long been overcome. We are called upon to take guidance from work on fluid dynamics (Lorenz, 1963), chaos mathematics (Gleick, 1988), and why not quantum theory (Busemeyer & Wang, 2015) or computational modeling? In fact, Busemeyer and Pothos (2012) showed in this journal that quantum probability modeling yields the same results for the effects of social projection on cooperation in social dilemmas as does the Bayesian modeling presented by the authors (Krueger et al., 2012). There has been more progress. Tarantola et al. (2017) used computational modeling to show how egocentric projection affects social learning and predictions about the behavior or others (see also Todd & Tamir, 2024).

We are all in favor of modeling complex systems parsimoniously, and as we noted above, we encourage theory-driven research on moderator variables. At the input level, for example, different types of social distance metrics can be fruitful for the exploration of the robustness of differential projection (Koch and colleagues). Likewise, the study of cultural differences is of abiding interest

(Kroupin et al., 2024). It would be interesting to see if the heightened ingroup favoritism in collectivist, as compared with individualist, cultures (Chen et al., 1998) can be explained by a differential combination of self-images and social projection, or by other, culture-bound, variations in cognition (Varnum et al., 2010).

Parsimony protects us from overfitting the data. Sedikides and Skowronski reflect on perceptions of similarity between a USA-based scholar and a Polish scholar, and how these perceptions may vary with recent experience such as a trip to Poland. The more fine-grained such examples are, they more they suggest the need for case studies as opposed to general theory. A general model runs the risk of overfitting the data. Our critics continue by noting that self-focused perceptions can vary from moment to moment. A general model treats such variation as noise (see Johnson). They further note that perceivers have detailed knowledge of specific others. This is true enough, but the IRM shields us from adding parameters to capture our perceptions of Uncle Leo.

In short, we think that Sedikides and Skowronski fail to distinguish between modeling complexity complexly and predicting more variance by adding parameters or moderator variables. The former is not as *du jour* as they seem to think. Maturana and Varela (1980) deserve credit for their pioneering work on complex self-organizing systems. The latter is a temptation the IRM was developed to curb. To make a point by adding a significant moderator variable is to advocate weak science; strong science removes unnecessary variables (Platt, 1964; Popper,1968). The generative power of the model can be expressed as the ratio of the number of ingoing assumptions (premises, axioms, independent variables) over the number of outgoing predictions. Here, the IRM is stronger than the ETM.

Sedikides and Skowronski also question the value of point predictions, ignoring the fact that the sort of complex modeling they admire may also yield point predictions. As to climate research, the prediction of greatest interest to humanity is how much the planet is going to heat up (within a *Spielraum*) over the lifetime of our children and grandchildren (IPCC, 2022). To note that some point-less predictions, that is, ordinal ones, can be of great value in well-designed experiments does not mean that precise point predictions do not make for stronger science *ceteris paribus*. The real beef, we suspect, lies with the IRM's non-motivational character. To be motivated to see oneself as better than others, one's group as better than other groups, and groups as being different is to *want* these things to

be the case. There is no wanting or "lusting" in the IRM. If motivation is to be privileged, we consider the ETM to be teleological. If the IRM calls on Newton, the ETM calls on Aristotle. To Aristotle, an apple falls to the ground because it wants to return to its natural place. Surely, humans want to do things or see things in a pleasing way. These wants can be proximal causes, but they are not necessary (Sapolsky, 2023).

What about the ETM? If anachronism is to be a serious charge and if comparative reasoning is to be so paramount, we'd expect to see the ETM (Sedikides et al., 2021) anchored in more recent theory and research than the IRM. This is not so. The ETM is rooted in Husserl's phenomenology, which was published in German around the turn of the 20th Century, and in James's (1890) pragmatism, which flourished at the same time. By comparison, the IRM's intellectual ancestor is the behaviorist and experimentalist Floyd Allport (1924). So, both models have deep intellectual roots, which we consider to be a credit. We also note that a plurality of references in the exposition of the ETM are from the 1990s and the aughts. We agree it is good practice to highlight the field's intellectual history, but we advise against the recycling of obsolete concepts such as the false consensus effect or the notion that self-enhancement equals error. Lastly, we are puzzled why the ETM does not incorporate the concepts and tools of dynamical-systems modeling, chaos theory, and fluid mechanics. If these paradigms are the key to Post-Newtonian psychological science, then why don't Sedikides and Skowronski model self-enhancement and self-protection with them?

The core of the ETM, as we understand it, is formed by its motivational assumptions. Most people have positive self-views, perceive themselves as similar to as well as superior to others because *they want to*. It makes them happy. These assumptions appear to be axiomatic and unfalsifiable. You believe them or you do not. We, in contrast, believe that cognitive processes are sufficient (and testable) to account for these positions. They give us baselines. Of course, something else might be going on, and researchers are invited to test and tell. The ETM places a shell of trivially true assumptions around its core, assumptions such as "An actor's behaviors often have many possible meanings" (p. 250). This must be hard to falsify!

Back to the future

We thank our commentators for a thoughtful and engaging discussion. Many of their suggestions are constructive and useful for the furthering of this research program. Some comments are challenging, and this is how it should be in a community of disputatious individuals. Given the number of comments made and points raised, we had to be selective in our response. We sought to focus on the most critical arguments, trying to avoid the mistake of ignoring criticism in hopes it might just go away. To this end, we selected the issues of model coherence, validity, sufficiency, and timeliness to organize our response.

In closing, we concede that we used the concept of paradigm shift (Kuhn, 1962) rather liberally, repeating a peccadillo seen frequently in psychological writing. If the Copernican revolution was a paradigm shift, the introduction of the IRM certainly is not. We can only hope that in a small way, the IRM will contribute to a paradigm *drift*, a re-orientation of research to the study of the interdependence of phenomena and outcomes. In that sense, the IRM is a plea for the study of complexity, while it is also a call to keep it simple whenever possible. We have worked with a minimum set of assumptions to ground the model, while striving to predict multiple outcomes with precision. While these efforts may not be sufficient to represent a paradigm, they reflect a *Denkstil*, or mode of thinking, as described by Ludwik Fleck (1935), whose ideas on how scientific findings become facts anticipated many of Kuhn's thoughts (Krueger, 2012). If psychological science lingers in the limbo of a pre-paradigmatic stage, as Kuhn asserted, there is no excuse for its practitioners to not cultivate a disciplined *Denkstil*. This being said, we strongly discourage researchers from submitting to a culture of conformity (Fiedler, 2024). Instead, we hope – with Shweder (epigraph) – that a community of disputatious individual is the most fertile ground for progress in any field.

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